From Prephysical Hypotheses to Physical Models

Program questions, mathematical models and physical correspondence

RESEARCH QUESTIONS

Specify

What primitives, relations and equivalences can be defined without assuming the physical structure being sought?

Derive and challenge

Which conclusions follow from stated assumptions, and which proposed implications fail?

Discriminate

Which observations distinguish candidate models when calibration and uncertainty are included?

PARALLEL MATHEMATICAL APPROACHES

Each row retains its own assumptions and scope.

APPROACH	MATHEMATICAL SETTING	STATEMENTS AND LIMITS
Signed-sector signal proposal ONE EXPLORATORY OPTION	Proposed sectors and a relation, without assigned physical frequency. Mathematical types remain to be specified.	Geometry and a continuum are proposed targets. The sketch supplies no construction or observation law.
Finite relational models CONDITIONAL MATHEMATICS	Finite positive-arity signature; nonempty finite structures; ordinary homomorphisms and homomorphic equivalence. →	Each equivalence class has a core unique up to isomorphism. Quotienting can discard multiplicity.
State/effect and instrument models MATHEMATICAL REPRESENTATIONS	Hilbert spaces, completely positive maps or stochastic kernels are mathematical choices with explicit observation rules.	Conditional output laws, instrument bounds and comparison models. Physical realization needs separate warrant.
Identification and information EXPERIMENT-RELATIVE ANALYSIS	Declared model class, experiment menu and outcome-law map; separate resource and statistical contracts. →	Identified properties are constant on observation fibers. Information and testing bounds keep their extra premises.
Analytic and effective models MODEL-SPECIFIC ANALYSIS	Chosen operators, transfer kernels, interaction laws and limiting regimes.	Spectral, entropy and continuum estimates retain their analytic hypotheses and unresolved extensions.

PHYSICAL CORRESPONDENCE: WHAT WOULD WARRANT A PHYSICAL INTERPRETATION?

Map the representation

Relate mathematical states, transformations and effects to preparations, controls and observations.

Justify physical quantities

Supply units, clocks, energy, geometry and composition where those physical claims are made.

Test discriminating predictions

State alternatives, calibration and uncertainty; obtain observations that distinguish the proposed models.

Arrows mark conditional mathematical implications. Grouping indicates a research topic, not causal propagation or physical emergence.

The figure organizes selected research lines; it does not assert one completed physical theory or a unique ontology.